

Theoretical Probability



1

a 17

b $\frac{12}{17}$

c 9

d 7

2

a $\frac{13}{20}$

b $\frac{7}{20}$

3

a $\frac{1}{2}$
e $\frac{9}{52}$

b $\frac{5}{13}$
f $\frac{7}{26}$

c $\frac{5}{26}$
g $\frac{1}{2}$

d $\frac{7}{13}$

4

a $\frac{1}{5}$

b $\frac{2}{5}$

c Yes

d No

Probability of complimentary events

5 0.947

6 0.64

7 0.57

8 0.44

9 0.032

10

a Drawing a number less than or equal to 61.

b $\frac{39}{100}$

11

a Drawing a number greater than or equal to 42.

b Drawing a number that is less than 43.

c $\frac{9}{50}$

12

a $\frac{25}{26}$
e $\frac{17}{26}$

b $\frac{25}{26}$
f $\frac{10}{13}$



c $\frac{23}{26}$
g $\frac{21}{26}$

d $\frac{23}{26}$
h $\frac{21}{26}$

13

a 0

b $\frac{5}{7}$

14

a $\frac{1}{2}$
e $\frac{5}{6}$
i 0

b $\frac{1}{2}$
f $\frac{2}{3}$

c No
g $\frac{1}{2}$

d No
h 1

15

a $\frac{23}{44}$

b
i $\frac{34}{69}$

ii $\frac{35}{69}$

16

a $\frac{37}{139}$

b $\frac{60}{139}$

c $\frac{102}{139}$

d $\frac{79}{139}$

17

a $\frac{3}{8}$

b $\frac{5}{8}$

c Yes

d Yes

18

a $\frac{4}{5}$
e $\frac{26}{45}$

b $\frac{7}{9}$

c $\frac{26}{45}$

d $\frac{382}{585}$

19

a $\frac{1}{8}$

b $\frac{7}{8}$

c $\frac{3}{8}$

d $\frac{5}{8}$